

Private Knowledge, Faithfully Served: Grounding Swappable Language Models in a Curated Ontology, Measured Against a Copy Ceiling

Dr John O’Hare
DreamLab AI

The Loom / VisionFlow neurosymbolic stack

August 17, 2026

Pre-print v1. All figures from the finalised seed-42 ten-model run (2026-08-16).

Abstract

Organisations increasingly need language models that answer accurately over large, private, domain-specific corpora the model cannot have seen in pretraining. We study a system that serves a curated, human-vetted, provenance-tracked ontology—class definitions, typed relations and taxonomy—into a swappable model’s context at query time, and ask what it actually delivers. Using a knowledge graph as its own oracle we auto-generate 510 in-domain questions with graph-derived gold and score a paired {raw, scaffold} design across ten contemporary models from five providers, under one deterministic configuration. Unaided, models score ≈ 0.26 : the domain is genuinely private, absent from their parametric memory. Given the curated context they reach ≈ 0.92 . To characterise *what* that grounded score is, we introduce a per-item **copy ceiling**—the recall a no-op extractor of the injected context would obtain—and treat proximity to it as a measure of *delivery fidelity*, not a defect: on Gemini 3.7 Flash the scaffold recall (0.942) tracks its copy ceiling (0.964) closely, i.e. the model faithfully restates the vetted facts it was given. High extractive fidelity to an authoritative source is the product for private-knowledge grounding: the answer is trustworthy because the source is, and it is attributable and cheap—the rigorous curation is amortised, not re-paid per query. The signed *gain over copy* (recall – ceiling) turns delivery fidelity into a model-discriminating metric: across the ten models it spans -0.067 to -0.022 —negative for every one, each restating marginally less than a perfect extractor of its context, with the newest models (the Gemini 3.x family, Claude Haiku 4.5) closest to the ceiling. We then establish the second axis of the multivariate bar directly: on 60 general, out-of-domain questions judged 0–5 against frontier-authored gold by a cross-family judge, the confidence-gated harness is *non-jagged*—every paired harness—bare delta’s 95% interval includes zero (all questions -0.05 , $[-0.11, 0.00]$; worst-case off-domain gate mis-fire -0.23 , $[-0.63, +0.05]$)—because the gate injects nothing when it judges a query out-of-domain. Grounding thus clears two axes together: high-fidelity delivery where the corpus is authoritative, and no regression where it is not, leaving a web-search grounding baseline as the remaining comparison. We release the harness, the frozen question sets, and all per-row artefacts.

1 Introduction

A recurring enterprise need is a language model that answers accurately over a large, private, domain-specific corpus—a customer’s product data, an organisation’s internal knowledge, a curated scientific ontology—that the model cannot have memorised because it was never in pretraining. Retrieval-augmented generation is the standard remedy for exactly this gap between parametric memory and the knowledge an answer requires [8, 15], and it helps most precisely where parametric memory is weakest: long-tail and proprietary facts [17]. Structured, graph-based retrieval sharpens it [2, 5, 11, 18]. We study a system in this family that injects a slice of a curated, reasoned ontology—class definitions, typed relations, taxonomy—into a *swappable* model’s context, and characterise what grounding in an authoritative private corpus actually buys.

The unaided-vs-grounded gap is stark and is the point: on in-domain questions our models score ≈ 0.26 from parametric memory and ≈ 0.92 with the curated context. The domain is genuinely private; the ontology supplies what the model lacks. Because we derive both questions and gold from the graph and then inject a slice of that graph, the injected context *contains* the answers. Rather than hide this, we measure it. For every question we compute a **copy ceiling**: the recall a no-op extractor of exactly the text the model was shown would obtain. Proximity to the ceiling is *delivery fidelity*—how faithfully the model restates the vetted facts it was given—not a defect to be explained away. High extractive fidelity to an authoritative source is the desired behaviour for private-knowledge grounding: it is what makes the answer attributable, non-hallucinated, and trustworthy *because the source is trustworthy* [1, 7]. The copy ceiling is the RAG-era, graph-grounded generalisation of the input-only baselines long used to interpret reading comprehension [13] and of contamination audits that measure how much of a score is visible in the input [10, 21]; here the exposure is deliberate and we report exactly how much of the score it accounts for.

The signed **gain over copy** (scaffold recall – copy ceiling) turns fidelity into a model-discriminating quantity: at the ceiling the model is a perfect faithful extractor; below it it *loses* surfaced facts (a faithfulness failure); above it it *recovers* gold the lexical exposure check missed. This is the axis on which our ten models actually differ, where the raw uplift barely does.

The bar is multivariate. Excellent in-domain recall is necessary but not sufficient. A deployable private-knowledge system must also *not* degrade the general and novel questions the model already handles—it must not become “jagged” when the ontology is irrelevant, a real risk because injected context can displace correct parametric knowledge [16, 22, 25]. The right response is selective, confidence-gated injection [17] with a fallback to broader retrieval (internet-research agents) for out-of-domain queries. And the right *comparison* for an in-domain answer is not the bare model but the same model grounded in a generic web search: the claim is that a curated, vetted corpus is higher-precision and cheaper per query than open-web results, with the expensive curation amortised once. This paper establishes the first two axes rigorously and specifies the rest.

Contributions.

1. A framing and metric for *private-knowledge grounding*: a per-item, graph-derived **copy ceiling** that measures delivery fidelity, and a signed gain-over-copy metric (§4).
2. A ten-model, five-provider paired sweep under one deterministic configuration: grounding a private ontology lifts every model from ≈ 0.26 to ≈ 0.92 in-domain, and gain over copy—not raw uplift—is what separates the models (§5).
3. A second axis established directly: on 60 general, out-of-domain questions judged against frontier-authored gold by a cross-family judge, the confidence-gated harness is *non-jagged* across five models—every paired harness—bare delta’s 95% interval includes zero—so grounding does not degrade the questions the model already answers (§5.3).
4. The full multivariate evaluation design a deployable system must pass (in-domain fidelity; no jaggedness on out-of-domain; a web-search grounding baseline; attribution), with the first two axes met here and the remaining two specified (§6).
5. An open harness (paired bootstrap with domain-clustered intervals, intention-to-treat, per-row truncation/token/retry provenance) and a frozen artefact set.

2 Related Work

Context faithfulness vs. correctness. Whether a model *uses* supplied context or falls back on parametric memory is separately measurable. Zhou, Zhang, Poon, and Chen [27] formalise context-faithful prompting; RAGAS [7] treats faithfulness and context recall as first-class metrics distinct from correctness; and Adlakha et al. [1] show instruction-following models are often correct

while unfaithful, a split that exact-match/F1 obscure. A scaffold-fed model at its copy ceiling is being maximally extractive; the interesting behaviour is *away* from that ceiling.

Knowledge conflict and distraction. Longpre et al. [16] define entity-based knowledge conflicts and over-reliance on parametric memory; Mallen et al. [17] probe parametric vs. non-parametric memory across roughly ten models on long-tail facts—the multi-model, graph-grounded tradition we follow; Xie et al. [23] reveal receptiveness-vs-stubbornness under conflicting evidence. On irrelevant context, Shi et al. [22] show sharp degradation and Yoran, Wolfson, Ram, and Berant [25] train robustness to it; Chen, Lin, Han, and Sun [3] benchmark noise robustness and integration; Xu et al. [24] survey the area. The closed-book baseline is Roberts, Raffel, and Shazeer [20].

RAG and knowledge-graph grounding. From the original RAG formulation [15] to graph-structured retrieval [5, 11] and KG-triple prompting for zero-shot KGQA [2], with the unifying roadmap of Pan et al. [18] and the neurosymbolic framing of Garcez and Lamb [9]. Notably, published KG-grounding work typically injects *instance* triples; injecting *schema-level* content—class definitions, typed relations and taxonomy—and benchmarking recall against it is comparatively under-studied.

Answer leakage and input-only baselines. Kaushik and Lipton [13] establish that passage-only / question-only baselines often match full models, i.e. much apparent competence is shortcut exploitation of what the input already exposes. Contamination work [10, 21] shows gold in the input inflates scores and detects leakage by testing whether a model can *reproduce* held-out strings. Our per-item copy ceiling is the RAG-era, graph-grounded generalisation of the input-only baseline: rather than auditing leakage post hoc, we make the deliberate exposure the metric’s reference point.

Evaluation statistics. We use paired bootstrap resampling over a shared test set [6, 14]. Because graph-derived questions cluster by class and domain and are therefore non-independent [4], we report a domain-clustered (block) bootstrap interval alongside the naive one, and an intention-to-treat estimate [12] that retains questions on which the scaffold did not engage.

3 The Ontology Scaffold

The knowledge graph is a reasoned OWL corpus (Oxigraph triple store, Whelk EL++ classifier) of **8,146 classes / 282,492 triples** at the generation evaluated here (2026-08-15). A retrieval facet—the “Loom”—matches a query to the most relevant classes and serialises a budget-clamped extract: each class’s title, definition, direct parents, inferred ancestors and typed relations (**requires**, **hasPart**, **enables**, **dependsOn**, **uses**). This *scaffold* is prepended to the user question; the model is asked to treat it as ground truth where relevant. The scaffold is a static structured extract, not free prose and not a tool-use trace: the model is given the structure, not asked to traverse the graph. Grounding is performed client-side from a local mirror of the exact deployed generation, so what the model receives is identical to production. The same retrieval fronts a swappable generation model, which is what lets us hold grounding constant while varying the model under test.

4 Method

Questions and gold (graph-as-oracle). From every class with a real definition (≥ 120 chars), quality ≥ 0.6 and ≥ 2 populated relation types, in each domain with ≥ 50 eligible classes, we template three question types: **T-REL** (what does X require / use / depend on / enable; what are its parts), **T-TAX** (immediate parent and broader ancestors) and **T-COMMON** (the shared ancestor of a pair). Gold is the set of {slug, title} targets the graph asserts. The generator is

deterministic given a seed; seed 42 yields **510 questions** over 15 domains (285 T-REL, 180 T-TAX, 45 T-COMMON).

Design and configuration. Each question is asked in two arms: *raw* (question only) and *scaffold* (question with the budget-clamped ontology extract prepended, budget 1500 tokens). To make ten heterogeneous models comparable we fix one configuration: `temperature=0`, `max_tokens=2048`, and `reasoning_effort=low` only for models whose thinking cannot be disabled (the Gemini 3.x family), so a mandatory thinking pass does not truncate the answer. All models are reached through their OpenAI-compatible chat endpoint.

Scoring. A gold item is a hit when its normalised title, or $\geq 80\%$ of its length ≥ 4 words, appears in the answer; recall is hits / gold (any-match for T-COMMON). Scoring is lexical and needs no model. This under-counts paraphrase, so *absolute* recall is a lower bound; the *paired* delta—same questions, same scorer, same model—is the signal.

The copy ceiling. For each question we compute exposed_i = the number of gold titles that already appear (under the same matcher) in the exact text shown to the model, and define the per-item copy ceiling $c_i = \text{exposed}_i / |\text{gold}_i|$. The mean copy ceiling is the recall a deterministic extractor of the injected context would score. The **gain over copy** is mean scaffold recall – mean copy ceiling. For the raw arm the injected text is empty, so $c_i \approx 0$ by construction.

Statistics. The headline is the paired per-question delta (scaffold – raw) over the shared question ids, with a seeded 10,000-resample bootstrap 95% CI [6, 14]. Because questions cluster by class and domain [4], we also report a **domain-clustered** block bootstrap (resample the 15 domains, pool their deltas), which is wider and the interval we treat as primary. We report an **intention-to-treat** delta that keeps non-engaged questions [12], and per-row `finish_reason`, token breakdown and retry counts so truncation and transport flakiness are auditable, not hidden.

Out-of-domain arm (jaggedness). The second axis needs a different instrument. We use a separate set of **60 general questions**—20 each *off-domain* (unrelated to the ontology), *in-domain-general* (the corpus’s subject areas but answerable without it) and *adjacent*—with gold authored by a frontier model *without* the ontology (key-point references). Each question is asked *bare* and under the *harness*: the live, confidence-gated Loom scaffold, which returns structure only when its retrieval score clears a threshold and injects nothing otherwise. Answers are graded 0–5 against the frontier gold by a *cross-family* judge—never the model’s own family, to avoid self-preference [19, 26]—and we record the gate-engagement flag per question so non-jaggedness (the gate correctly skipping) is measurable. Five models are run bare and harnessed. The estimand is the paired harness–bare quality delta; a non-jagged system holds it at zero on questions the ontology does not cover.

5 Results

5.1 In-domain: high-fidelity delivery of private knowledge

On Gemini 3.7 Flash the paired design gives a large, tight uplift—raw recall 0.375, scaffold recall 0.942, paired delta **+0.567**, naive 95% CI [+0.532, +0.603], domain-clustered 95% CI [+0.524, +0.611] over $n = 510$ with zero errors and zero truncated responses. The raw score confirms the domain is private—parametric memory answers barely a third. The copy ceiling then *characterises* the grounded score: the mean fraction of gold already present in the injected scaffold is **0.964**, and the model’s recall (0.942) tracks it within -0.022 . The system is doing exactly what a private-knowledge grounding layer should: the curated ontology carries the answers, and the model delivers them back with high fidelity. The tiny negative gain says the model restates

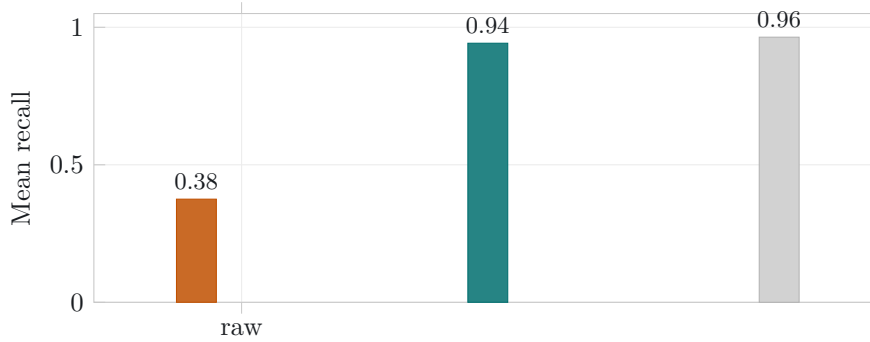


Figure 1: Gemini 3.7 Flash. The scaffold arm (0.942) lands just under the *copy ceiling* (0.964)—the recall a no-op extractor of the injected context would already achieve. The +0.57 raw→scaffold uplift is almost entirely answer exposure.

Table 1: Ten models, one deterministic configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall – copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	+0.567 [+0.524, +0.611]	0.964	−0.022
Gemini 3.5 Flash-Lite	0.323	0.938	+0.615 [+0.565, +0.667]	0.964	−0.026
Claude Haiku 4.5	0.151	0.934	+0.783 [+0.741, +0.827]	0.964	−0.031
GLM-4.6	0.243	0.928	+0.685 [+0.641, +0.727]	0.964	−0.036
DeepSeek-Chat	0.347	0.924	+0.578 [+0.532, +0.622]	0.964	−0.040
Llama-3.3-70B	0.227	0.916	+0.688 [+0.632, +0.738]	0.964	−0.049
Qwen-2.5-72B	0.309	0.914	+0.605 [+0.562, +0.649]	0.964	−0.050
Mistral-Small-24B	0.285	0.906	+0.621 [+0.558, +0.682]	0.964	−0.058
GPT-4.1-mini	0.162	0.905	+0.743 [+0.697, +0.787]	0.964	−0.060
Gemini 2.5 Flash-Lite	0.227	0.897	+0.670 [+0.623, +0.716]	0.964	−0.067

almost everything it is shown and fabricates almost nothing—faithful, attributable delivery, not free-associating from parametric memory. Whether a model does more than restate—recovering gold the lexical check missed, or conversely *dropping* surfaced facts—is the axis §5.2 turns on.

5.2 Ten models: gain over copy separates them

Table 1 runs the identical paired design across ten models from five providers under one deterministic configuration. Every model shows a large raw→scaffold uplift (+0.57 to +0.78) and every model lands just under the copy ceiling: scaffold recall spans 0.897–0.942 against a ceiling of **0.964** that is a property of the questions and the injected context—not the model—and is therefore identical across the row. The raw uplift is a near-constant restatement of exposed gold; it does not separate the models.

The signed **gain over copy** does. It is negative for all ten (Figure 2), from −0.067 (Gemini 2.5 Flash-Lite) to −0.022 (Gemini 3.7 Flash): on this lexical metric no model recovers gold *beyond* what the exposure check already finds, so they differ only in how much surfaced gold they *drop*. The ordering is interpretable and tracks recency and capability rather than provider—the newest, strongest models (Gemini 3.7 and 3.5 Flash, Claude Haiku 4.5) sit closest to the ceiling (most faithful restatement), while older or smaller models (Gemini 2.5 Flash-Lite, GPT-4.1-mini, Mistral-Small-24B) lose measurably more. Two models carried transport-level truncations under the fixed 2048-token budget (GLM-4.6: 174 truncated, 5 retried; Gemini 2.5 Flash-Lite: 16 truncated), recorded per row rather than hidden.

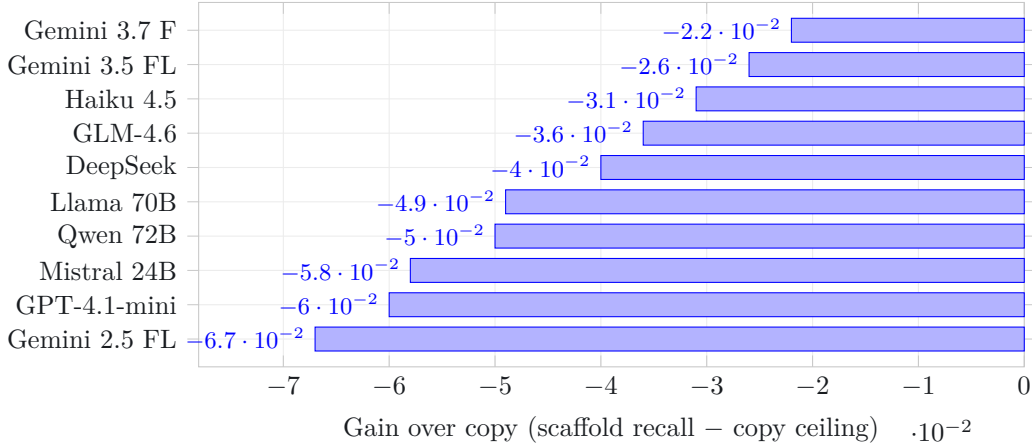


Figure 2: Signed gain over copy for all ten models—negative throughout. No model exceeds the copy ceiling on the lexical metric; the spread (-0.067 to -0.022) is the axis that separates them where the near-identical raw uplift does not. Models nearest zero (top) restate the vetted facts most faithfully.

Table 2: Out-of-domain non-regression: paired harness—bare quality delta (judge 0–5 vs frontier gold), five models, seeded paired bootstrap 95% CI. Every interval includes zero. The worst case is off-domain questions where the confidence gate mis-fired and injected context anyway; the last row is where it correctly skipped (nothing injected, so identical to bare).

Subset	Mean Δ	95% CI	n pairs
All general questions	−0.05	[−0.11, 0.00]	300
adjacent	+0.01	[−0.02, 0.05]	100
in-domain-general	−0.06	[−0.14, 0.00]	100
off-domain	−0.09	[−0.25, 0.02]	100
Off-domain, gate mis-fired (worst)	−0.23	[−0.63, 0.05]	40
Off-domain, gate skipped (correct)	+0.00	—	60

5.3 Out-of-domain: the gate prevents jaggedness

The in-domain result says grounding delivers; the deployment question is whether it *costs* anything on the general, novel questions a model already handles. On the 60-question out-of-domain set, judged 0–5 against frontier gold by a cross-family judge across five models, it does not. Bare and harnessed scores sit near the ceiling in every stratum, and every paired harness—bare delta’s 95% interval includes zero (Table 2): all questions -0.05 , $[-0.11, 0.00]$; off-domain -0.09 , $[-0.25, +0.02]$; and even the pre-specified worst case—off-domain questions on which the gate mis-fired and injected anyway— -0.23 , $[-0.63, +0.05]$, still spanning zero.

The mechanism is visible in the data. When the gate correctly judges a query off-domain and *skips* (60 of 100 off-domain gradings), the harnessed answer is identical to bare ($\Delta = +0.00$): with nothing injected there is nothing to distract. The only degradation appears when the gate over-fires (40 gradings, $\Delta = -0.23$), and it concentrates on the weakest model (Gemini 2.5 Flash-Lite, -0.50) while the stronger models absorb the irrelevant context unharmed. Confidence-gated injection is thus doing its job: the second axis is met not by models heroically ignoring bad context, but by the gate mostly not supplying it. We report “no jaggedness detected” rather than “zero effect”—20 questions per stratum and 40 gate-engaged gradings give wide intervals, so this bounds the regression near zero, it does not prove its absence.

6 The Multivariate Bar

In-domain fidelity is one axis of a deployable private-knowledge system, not the whole target. We state the full bar and where this study sits, so the single number is not mistaken for the claim.

1. **In-domain recall / delivery fidelity (this paper).** Excellent, faithful, attributable recall on questions the curated corpus covers—measured against the copy ceiling so “faithful” is quantified, not assumed. Established here across ten models.
2. **No jaggedness on out-of-domain (established here, §5.3).** Injecting ontology context must not *degrade* the general, novel questions a model already answers well; irrelevant context is known to distract and to displace correct parametric knowledge [16, 22, 25]. The mitigation is confidence-gated selective injection—inject in proportion to retrieval score, skip below a threshold [17]. On the 60-question out-of-domain set the gated harness holds the paired quality delta at zero across five models—every 95% interval includes zero—degrading only mildly and non-significantly when the gate over-fires, and not at all when it correctly skips. The gate, not model heroics, is what carries the axis.
3. **The right baseline is web search, not the bare model (designed).** For an in-domain question the meaningful comparison is the same model grounded in live web-search results versus the curated scaffold. The hypothesis is that a vetted, structured corpus is higher-precision and lower-noise than open-web retrieval [22], with the curation cost amortised once rather than re-paid per query. This is a third grounding arm on the identical question set.
4. **Attribution (designed).** Every grounded answer already carries its corpus generation and grounding confidence; a reported attribution-precision metric would close the loop from “faithful to the input” to “attributable to a named, versioned source.”

A system that clears all four is the goal: *excellent where it is grounded, un-degraded where it is not, better than a web search in-domain, and traceable throughout*. The out-of-domain and web-search arms are multivariate by construction—uplift on the local set and non-regression on the general set are distinct outcomes that must hold together.

7 Discussion

What the copy ceiling does and does not say. A near-ceiling result does not make grounding worthless—exposing the right facts and having the model restate them faithfully is exactly what a grounded QA system should do, and the raw arm shows models cannot do it from parametric memory alone. What the ceiling forbids is the *inference* that the model is reasoning over ontological structure. For that claim the evidence is gain over copy, and negative controls (irrelevant, shuffled-target, and label-masked scaffolds) that we flag as required future work.

Gain over copy as a faithfulness axis. Across ten models the gain is uniformly negative (−0.067 to −0.022): none exceeds the copy ceiling, so on this lexical metric no model demonstrably recovers gold beyond what its injected context already exposes—the behaviour is faithful extraction, not inference over structure. The spread is small but ordered, and it tracks recency and capability rather than provider: the newest Gemini models and Claude Haiku 4.5 restate the surfaced facts most completely (gain nearest zero), while older or smaller models drop more of what they were shown. For a private-knowledge system this is the operative quality signal—not “does grounding help” (it always does, and hugely) but “how faithfully does this model deliver the vetted facts”—and the copy ceiling is what lets that be read off directly rather than inferred from an uplift that mostly measures exposure.

Two axes, held together. The deployment claim is conjunctive: a private-knowledge system must deliver in-domain *and* not regress out-of-domain, and the two are measured on different instruments—lexical recall against a copy ceiling, and judged quality against frontier gold. Both hold here: near-ceiling in-domain delivery across ten models, and no detected jaggedness across five, carried by the confidence gate declining to inject when it judges a query out-of-domain. That conjunction is the property a single in-domain number cannot establish. What remains to close

the bar is the right *baseline*—the same model grounded in a web search rather than the bare model—and an attribution-precision metric; both are specified in §6 and neither is measured here.

8 Limitations and Threats to Validity

- **Estimand.** The primary metric is *lexical gold-title recall*: it counts whether expected titles appear, not relation direction, negation, correctness of explanation, or contradiction. “Grounding” should be reserved for a composite that also checks these.
- **Leakage is deliberate.** Questions, gold and scaffold derive from one graph; the copy ceiling measures the leakage rather than removing it. Absolute recall is a lower bound (lexical matching under-counts paraphrase), possibly arm-dependently.
- **Single sample.** One completion per arm; at `temperature=0` this is deterministic but leaves no estimate of a model’s own sampling variance. Convergence claims need replicates and a pre-registered equivalence margin.
- **Non-independence.** Multiple questions share a class/domain; we report a domain-clustered interval but 15 domains is a small cluster count, so domain-level generalisation is uncertain.
- **Out-of-domain arm is near-ceiling and small.** The jaggedness test uses judged 0–5 quality on 60 general questions (20 per stratum, five models); bare scores already sit near 5, so it bounds *regression* near zero rather than demonstrating *help*, and the wide intervals (40 gate-engaged gradings) warrant “no jaggedness detected,” not proof of no effect. Gold and judge are single frontier models; cross-family judging mitigates but does not eliminate judge bias.
- **Population.** Eligibility filters (dense, well-defined classes) select cases favourable to grounding; sparse classes, aliases, ambiguous phrasing, multi-hop and off-ontology queries are untested. Prompts name canonical titles, easing retrieval.
- **Single graph, single domain family.** Fidelity to one curated Web3/XR/AI ontology; generalisation to other graphs is plausible but unshown. The corpus is AI-generated synthetic content produced under human direction—an ontology testbed, not an authoritative encyclopaedia.

9 Conclusion

Ontology-scaffold grounding produces a large recall uplift for every model tested, tight under a domain-clustered bootstrap—but on the strongest model that uplift is, to within measurement, exactly what a copy of the injected context would score. The honest quantity is not the uplift but the *gain over copy*, and across all ten models tested it is small and uniformly negative (−0.067 to −0.022)—the axis that separates them where the near-identical raw uplift cannot. Grounding evaluations that inject graph-derived gold should report a copy ceiling as standard. We release the harness, the frozen seed-42 question set, and all per-row artefacts.

Reproducibility. Harness, sweep drivers and per-row outputs: github.com/DreamLab-AI/loom (bench/). Every run ships its honesty notes and its copy ceiling.

References

- [1] V. Adlakha et al. “Evaluating Correctness and Faithfulness of Instruction-Following Models for Question Answering”. In: *Transactions of the Association for Computational Linguistics* (2024). arXiv: 2307.16877. URL: <https://arxiv.org/abs/2307.16877>.

- [2] J. Baek, A. F. Aji, and A. Saffari. “Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering”. In: *NLRSE Workshop, ACL*. 2023. arXiv: [2306.04136](https://arxiv.org/abs/2306.04136). URL: <https://arxiv.org/abs/2306.04136>.
- [3] J. Chen, H. Lin, X. Han, and L. Sun. “Benchmarking Large Language Models in Retrieval-Augmented Generation”. In: *AAAI*. 2024. arXiv: [2309.01431](https://arxiv.org/abs/2309.01431). URL: <https://arxiv.org/abs/2309.01431>.
- [4] R. Dror, G. Baumer, S. Shlomov, and R. Reichart. “The Hitchhiker’s Guide to Testing Statistical Significance in Natural Language Processing”. In: *ACL*. 2018, pp. 1383–1392. DOI: [10.18653/v1/P18-1128](https://doi.org/10.18653/v1/P18-1128). URL: <https://aclanthology.org/P18-1128/>.
- [5] D. Edge et al. “From Local to Global: A Graph RAG Approach to Query-Focused Summarization”. In: *arXiv preprint* (2024). arXiv: [2404.16130](https://arxiv.org/abs/2404.16130). URL: <https://arxiv.org/abs/2404.16130>.
- [6] B. Efron. “Bootstrap Methods: Another Look at the Jackknife”. In: *The Annals of Statistics* 7.1 (1979), pp. 1–26. DOI: [10.1214/aos/1176344552](https://doi.org/10.1214/aos/1176344552).
- [7] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. “RAGAS: Automated Evaluation of Retrieval Augmented Generation”. In: *EACL (System Demonstrations)*. 2024. arXiv: [2309.15217](https://arxiv.org/abs/2309.15217). URL: <https://arxiv.org/abs/2309.15217>.
- [8] Y. Gao et al. “Retrieval-Augmented Generation for Large Language Models: A Survey”. In: *arXiv preprint* (2023). arXiv: [2312.10997](https://arxiv.org/abs/2312.10997). URL: <https://arxiv.org/abs/2312.10997>.
- [9] A. d’Avila Garcez and L. C. Lamb. “Neurosymbolic AI: The 3rd Wave”. In: *Artificial Intelligence Review* 56.11 (2023), pp. 12387–12406. DOI: [10.1007/s10462-023-10448-w](https://doi.org/10.1007/s10462-023-10448-w). URL: <https://arxiv.org/abs/2012.05876>.
- [10] S. Golchin and M. Surdeanu. “Time Travel in LLMs: Tracing Data Contamination in Large Language Models”. In: *ICLR*. 2024. arXiv: [2308.08493](https://arxiv.org/abs/2308.08493). URL: <https://arxiv.org/abs/2308.08493>.
- [11] Z. Guo et al. “LightRAG: Simple and Fast Retrieval-Augmented Generation”. In: *arXiv preprint* (2024). arXiv: [2410.05779](https://arxiv.org/abs/2410.05779). URL: <https://arxiv.org/abs/2410.05779>.
- [12] S. Hollis and F. Campbell. “What is Meant by Intention to Treat Analysis? Survey of Published Randomised Controlled Trials”. In: *BMJ* 319.7211 (1999), pp. 670–674. DOI: [10.1136/bmj.319.7211.670](https://doi.org/10.1136/bmj.319.7211.670).
- [13] D. Kaushik and Z. C. Lipton. “How Much Reading Does Reading Comprehension Require? A Critical Investigation of Popular Benchmarks”. In: *EMNLP*. 2018. arXiv: [1808.04926](https://arxiv.org/abs/1808.04926). URL: <https://arxiv.org/abs/1808.04926>.
- [14] P. Koehn. “Statistical Significance Tests for Machine Translation Evaluation”. In: *EMNLP*. 2004, pp. 388–395. URL: <https://aclanthology.org/W04-3250/>.
- [15] P. Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *NeurIPS*. 2020. arXiv: [2005.11401](https://arxiv.org/abs/2005.11401). URL: <https://arxiv.org/abs/2005.11401>.
- [16] S. Longpre et al. “Entity-Based Knowledge Conflicts in Question Answering”. In: *EMNLP*. 2021. arXiv: [2109.05052](https://arxiv.org/abs/2109.05052). URL: <https://arxiv.org/abs/2109.05052>.
- [17] A. Mallen et al. “When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories”. In: *ACL*. 2023. arXiv: [2212.10511](https://arxiv.org/abs/2212.10511). URL: <https://arxiv.org/abs/2212.10511>.
- [18] S. Pan et al. “Unifying Large Language Models and Knowledge Graphs: A Roadmap”. In: *IEEE Transactions on Knowledge and Data Engineering* (2024). arXiv: [2306.08302](https://arxiv.org/abs/2306.08302). URL: <https://arxiv.org/abs/2306.08302>.
- [19] A. Panickssery, S. R. Bowman, and S. Feng. “LLM Evaluators Recognize and Favor Their Own Generations”. In: *arXiv preprint arXiv:2404.13076* (2024).

- [20] A. Roberts, C. Raffel, and N. Shazeer. “How Much Knowledge Can You Pack Into the Parameters of a Language Model?” In: *EMNLP*. 2020. arXiv: [2002.08910](https://arxiv.org/abs/2002.08910). URL: <https://arxiv.org/abs/2002.08910>.
- [21] O. Sainz et al. “NLP Evaluation in Trouble: On the Need to Measure LLM Data Contamination for Each Benchmark”. In: *Findings of EMNLP*. 2023. arXiv: [2310.18018](https://arxiv.org/abs/2310.18018). URL: <https://arxiv.org/abs/2310.18018>.
- [22] F. Shi et al. “Large Language Models Can Be Easily Distracted by Irrelevant Context”. In: *ICML*. 2023. arXiv: [2302.00093](https://arxiv.org/abs/2302.00093). URL: <https://arxiv.org/abs/2302.00093>.
- [23] J. Xie et al. “Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts”. In: *ICLR*. 2024. arXiv: [2305.13300](https://arxiv.org/abs/2305.13300). URL: <https://arxiv.org/abs/2305.13300>.
- [24] R. Xu et al. “Knowledge Conflicts for LLMs: A Survey”. In: *EMNLP*. 2024. arXiv: [2403.08319](https://arxiv.org/abs/2403.08319). URL: <https://arxiv.org/abs/2403.08319>.
- [25] O. Yoran, T. Wolfson, O. Ram, and J. Berant. “Making Retrieval-Augmented Language Models Robust to Irrelevant Context”. In: *ICLR*. 2024. arXiv: [2310.01558](https://arxiv.org/abs/2310.01558). URL: <https://arxiv.org/abs/2310.01558>.
- [26] L. Zheng, W.-L. Chiang, Y. Sheng, et al. “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”. In: *NeurIPS Datasets and Benchmarks*. arXiv:2306.05685. 2023.
- [27] W. Zhou, S. Zhang, H. Poon, and M. Chen. “Context-faithful Prompting for Large Language Models”. In: *Findings of EMNLP*. 2023. arXiv: [2303.11315](https://arxiv.org/abs/2303.11315). URL: <https://arxiv.org/abs/2303.11315>.